Synthetic Citation Practices in Automated Bibliography Extraction

M. A. Datapoint, R. K. Groundtruth, and T. Weierstrass
Institute of Synthetic Bibliometrics, University of Nowhere

The measurement of scholarly impact has long depended on the careful parsing of printed bibliographies. Early approaches relied on manual transcription, a process both laborious and prone to systematic error. Subsequent work demonstrated that citation contexts carry substantial rhetorical information about the citing work (Hartwell and Ng, 1998). A recurring finding is that reference lists exhibit remarkable regularity within a journal but striking diversity across disciplines. Extraction pipelines must therefore tolerate variation in punctuation, ordering, and typographic convention. Several studies have shown that author name disambiguation remains the principal bottleneck in large scale indexing (Kjærgaard et al., 1999). The problem is compounded by initials, diacritics, and institutional authors that defy naive parsing rules. Optical character recognition introduces further noise, particularly in documents scanned from bound volumes. Typographic ligatures and discretionary hyphens are frequent sources of spurious tokens in extracted text (Vasquez and Takeda, 2000). Cross-column reading order errors account for a significant fraction of failures in two column layouts. Page furniture such as running heads and folios is routinely mistaken for bibliographic content. Robust segmenters exploit whitespace geometry rather than relying on lexical cues alone (Almeida, 2001). Supervised sequence labelling has emerged as the dominant paradigm for reference string parsing. Feature based models remain competitive when training data is scarce or domain shifted. The availability of style aware synthetic corpora has materially improved evaluation practice (Ostrowski et al., 2002). Ground truth construction, however, continues to demand painstaking manual verification. Digital object identifiers provide a strong anchor when present, yet coverage is far from uniform. Older literature is disproportionately affected by missing or malformed persistent identifiers (Moreau and Petrov, 2003). Conference proceedings and theses occupy a grey zone in many citation style definitions. Preprint servers have introduced new conventions that established styles accommodate only awkwardly. The interplay between volume, issue, and pagination fields is a perennial source of ambiguity (Fitzgerald et al., 2004). Renderer differences between processor implementations further complicate cross platform comparison. Evaluation protocols should therefore report style level metrics rather than a single aggregate score. Error analysis reveals that truncation of long author lists is mishandled by a surprising number of systems (Novak, 2005). The et alii device in particular interacts badly with naive regular expression approaches. Title casing conventions differ between styles and must be normalised before string comparison. Accented characters survive some extraction pipelines and are silently normalised away by others (Barakat and Johansson, 2006). A conservative normalisation strategy preserves the grapheme inventory of the source document. Downstream applications such as recommender systems are sensitive to even small error rates. Bibliographic coupling analyses assume near perfect recall of the underlying reference lists (Martinez-Campos et al., 2007). The methods section of this paper details a fully synthetic yet typographically faithful corpus. Each document was typeset programmatically under controlled layout perturbations. Imperfections were injected only where they plausibly arise in real production scanning workflows (Ashworth and Lindgren, 2008). The corpus spans author date and numeric citation traditions across ten widely used styles. All items are fictional and any resemblance to real publications is entirely coincidental. We release the corpus together with the generation scripts to encourage replication and extension (Grant, 2009).

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